



IITG RACING

Indian Institute of Technology Guwahati

Design and Development of Hydraulic Braking System for Formula Student Electric Vehicle

Mohnish Raja • Bedabrat Barma • Shirshendra Nath Saha • Manit Mehta

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1 Introduction

The braking system of a Formula Student Electric Vehicle (FSEV) is one of the most critical subsystems, governing the vehicle's ability to decelerate in a controlled and repeatable manner. A well-designed hydraulic disc braking system must balance multiple competing demands — delivering sufficient clamping force to lock all four wheels under maximum deceleration, while remaining compact, lightweight, and fully compliant with Formula Bharat and Formula Student rulebook specifications.

This report documents the end-to-end design and development of the hydraulic braking system for IITG Racing's FSEV. The work encompasses the selection of brake calipers, braking force calculations, brake line pressure analysis, pedal ratio optimisation, and thermal and structural FEA of brake disc and brake pedal. The system adopts a dual-circuit hydraulic layout with separate front and rear master cylinders, enabling independent adjustment of the brake bias ratio in accordance with the dynamic weight transfer characteristics of the vehicle.

All calculations are grounded in the vehicle's confirmed mass properties: a total vehicle-plus-driver mass of 305 kg, a wheelbase of 1590 mm, a centre-of-gravity height of 300 mm, and front/rear weight split optimised for the FSEV's packaging constraints.

2 Brake System Overview

2.1 System Description

The hydraulic brake system for the IITG Racing FSEV incorporates the following major subsystems:

- Dual-circuit hydraulic system — independent front and rear master cylinders
- Four-wheel disc brakes — two front and two rear calipers
- Wilwood GP200 floating calipers — selected after a comprehensive comparative analysis
- Brake rotors fitted inside 11-inch (279.4 mm OD) rims — maximum permissible rotor diameter: 210 mm

- Adjustable brake bias bar — allows in-cockpit front/rear pressure split tuning

2.2 Braking Principle

The hydraulic disc brake system operates on Pascal's Law: pressure applied at the master cylinder is transmitted equally in all directions through the brake fluid, actuating the caliper pistons simultaneously at each wheel. The resulting friction between the brake pads and the rotating disc generates a retarding torque that decelerates the vehicle.

When the driver applies a pedal force F_{pedal} , the pedal ratio (6:1) amplifies this force to produce a master cylinder output force. This creates a hydraulic pressure that propagates through the brake lines to the caliper pistons, which squeeze the pads against the disc, generating a braking torque proportional to the clamping force, pad friction coefficient, and effective disc radius.

3 Caliper Selection

3.1 Selection Criteria

The caliper selection was driven by the following engineering and regulatory requirements:

- Sufficient piston area to deliver the required clamping force at acceptable line pressures
- Minimum weight — critical for unsprung mass reduction and overall vehicle dynamics
- Compatibility with the rotor diameter constraint (max 210 mm OD) within the 11-inch rim
- Availability and FSAE precedent — proven performance in competition environments
- Caliper clearance: 30 – 40 mm within the rim envelope

3.2 Comparative Caliper Matrix

A systematic comparison of available calipers from AP Racing and Wilwood was conducted. Key parameters evaluated are summarised in Table 1.

Manufacturer	Model	Weight (lbs)	Disk OD (in)	Piston Area (in²)	# Pistons
AP Racing	Lug Mount CP2576	2.4	10.51	2.08	2
AP Racing	Lug Mount CP2577	2.4	10.51	2.41	2
AP Racing	Lug Mount CP3176	2.4	10.51	1.77	2
AP Racing	Lug Mount CP3177	2.4	10.51	1.58	2
AP Racing	Lug Mount CP3178	2.4	10.51	1.23	2
Wilwood	Dynapro Single IIIA	1.8	13.0	2.40	2
Wilwood	Billet GP 200	0.9	11.0	1.23	2
Wilwood	Dynapro Single	2.3	13.0	2.40	2
Wilwood	Dynapro Lugmount	3.8	13.0	4.80	4

Table 1: Comparative Caliper Matrix (selected caliper highlighted)

3.3 Selected Caliper — Wilwood GP200

The Wilwood Billet GP200 was selected as the optimal caliper for all four corners of the FSEV. Detailed justification is provided below.

Parameter	Value
Model	Wilwood Billet GP200
Type	2-Piston Floating Caliper

Parameter	Value
Body Material	Billet Aluminium
Piston Material	Stainless Steel
Weight	0.9 lbs (0.41 kg)
Disk OD Compatibility	11 inches (279.4 mm)
Disk Thickness	0.25 inches (6.35 mm)
Piston Area	1.23 in ² = 793.547 mm ²
Number of Pistons	2

Table 2: Wilwood GP200 Specifications

3.4 Justification for Selection

- **Lightweight:** At only 0.9 lbs, the GP200 is significantly lighter than competing options (AP Racing calipers weigh 2.4 lbs). This reduces unsprung mass by approximately 1.5 lbs per corner — a measurable benefit to vehicle dynamics.
- **Sufficient piston area:** The 1.23 in² piston area provides adequate clamping force for the required braking torque at line pressures within the system's operating range.
- **Compact packaging:** The caliper fits within the 30–40 mm clearance envelope available inside the 11-inch rim, compatible with the maximum 210 mm rotor diameter.
- **FSAE proven:** The GP200 is widely used in Formula Student competition, ensuring documented performance and readily available pad compounds.
- **Material advantage:** Billet aluminium construction provides high thermal conductivity for effective heat dissipation during repeated braking events on circuit.

4 Brake Force Calculations

4.1 Vehicle Parameters

Parameter	Symbol	Value	Unit
Vehicle + Driver Mass	m	305	kg
Wheelbase	L	1590	mm
CG Height	h_cg	300	mm
CG to Front Axle	l_f	923.5	mm
CG to Rear Axle	l_r	666.5	mm
Wheel Radius	R_w	228.6	mm
Tyre–Road Friction Coeff.	μ	0.70	—
Pad Friction Coefficient	μ_{pad}	0.43	—
Rotor Outer Radius	r_out	92.5	mm
Rotor Inner Radius	r_in	30.0	mm
Caliper Piston Area	A_cp	793.547	mm ²
Master Cylinder Bore Area	A_bore	387.913	mm ²
Pedal Ratio	P_r	6	:1
Initial Speed	v ₀	120	km/h (33.33 m/s)

Table 3: Vehicle and System Parameters

4.2 Static Axle Load Distribution

Without braking, the static load on each axle is calculated from the vehicle's weight and CG position:

$$\begin{aligned}W_{\text{total}} &= m \times g = 305 \times 9.81 = 2992.1 \text{ N} \\W_{\text{f,static}} &= W_{\text{total}} \times l_r / L = 2992.1 \times 0.6665 / 1.59 = 1254.2 \text{ N} \\W_{\text{r,static}} &= W_{\text{total}} \times l_f / L = 2992.1 \times 0.9235 / 1.59 = 1737.8 \text{ N}\end{aligned}$$

4.3 Dynamic Axle Loads Under Maximum Braking

During maximum braking, weight transfers forward. The deceleration and forward load transfer are:

$$\begin{aligned}a_{\text{max}} &= \mu \times g = 0.7 \times 9.81 = 6.867 \text{ m/s}^2 (\approx 0.7 \text{ g}) \\ \Delta W &= (h_{\text{cg}} / L) \times m \times a_{\text{max}} = (0.3 / 1.59) \times 305 \times 6.867 = 395.2 \text{ N} \\ W_{\text{f,dynamic}} &= W_{\text{f,static}} + \Delta W = 1254.2 + 395.2 = 1649.4 \text{ N} \\ W_{\text{r,dynamic}} &= W_{\text{r,static}} - \Delta W = 1737.8 - 395.2 = 1342.7 \text{ N}\end{aligned}$$

4.4 Brake Force Distribution

The total maximum braking force (friction-limited) and its distribution across the axles:

$$\begin{aligned}F_{\text{brake,max}} &= \mu \times W_{\text{total}} = 0.7 \times 2992.1 = 2094.4 \text{ N} \\ F_{\text{front}} &= (l_r / L) \times F_{\text{brake,max}} = (0.6665 / 1.59) \times 2094.4 = 878.0 \text{ N} \\ F_{\text{rear}} &= (l_f / L) \times F_{\text{brake,max}} = (0.9235 / 1.59) \times 2094.4 = 1216.5 \text{ N}\end{aligned}$$

4.5 Brake Torque Per Wheel

Dividing each axle's brake force equally between the two wheels and multiplying by the rolling radius:

$$\begin{aligned}T_f &= (F_{\text{front}} / 2) \times R_w = (878.0 / 2) \times 0.2286 = 100.4 \text{ Nm per wheel} \\ T_r &= (F_{\text{rear}} / 2) \times R_w = (1216.5 / 2) \times 0.2286 = 139.0 \text{ Nm per wheel}\end{aligned}$$

4.6 Effective Disc Radius

Using the standard annular pad geometry, the effective (mean) braking radius of the disc is:

$$\begin{aligned}r_{\text{eff}} &= (2/3) \times (r_{\text{out}}^3 - r_{\text{in}}^3) / (r_{\text{out}}^2 - r_{\text{in}}^2) \\ r_{\text{eff}} &= (2/3) \times (0.0925^3 - 0.030^3) / (0.0925^2 - 0.030^2) = 66.6 \text{ mm}\end{aligned}$$

4.7 Caliper Clamp Force

The clamp force required from each caliper (both pads acting) to generate the necessary brake torque:

$$\begin{aligned}F_{\text{clamp}} &= T / (2 \times \mu_{\text{pad}} \times r_{\text{eff}}) \\ F_{\text{fc}} &= 100.4 / (2 \times 0.43 \times 0.0666) = 1753 \text{ N (front)} \\ F_{\text{rc}} &= 139.0 / (2 \times 0.43 \times 0.0666) = 2429 \text{ N (rear)}\end{aligned}$$

4.8 Hydraulic Line Pressures

The brake line pressures required to generate the above clamp forces with the Wilwood GP200 ($A_{\text{cp}} = 793.547 \text{ mm}^2$):

$$\begin{aligned}P_{\text{front}} &= F_{\text{fc}} / A_{\text{cp}} = 1753 / 793.547 \times 10^{-6} = 2209 \text{ kPa} \approx 22.1 \text{ bar} \\ P_{\text{rear}} &= F_{\text{rc}} / A_{\text{cp}} = 2429 / 793.547 \times 10^{-6} = 3061 \text{ kPa} \approx 30.6 \text{ bar} \\ F_{\text{mc}} &= (P_{\text{front}} + P_{\text{rear}}) \times A_{\text{bore}} = (2209 + 3061) \times 10^3 \times 387.913 \times 10^{-6} = 2044 \text{ N} \\ \text{Brake Bias (front)} &= P_{\text{front}} \times A_{\text{bore}} / F_{\text{mc}} = 0.42 \text{ (42\%)} \\ F_{\text{pedal}} &= F_{\text{mc}} / P_r = 2044 / 6 = 340.7 \text{ N}\end{aligned}$$

4.9 Stopping Distance

The theoretical stopping distance from 120 km/h (33.33 m/s) under maximum friction-limited deceleration:

$$d_{\text{stop}} = v_0^2 / (2 \times a_{\text{max}}) = 33.33^2 / (2 \times 6.867) = 80.9 \text{ m}$$

Parameter	Value	Unit
Total Vehicle Weight	2992.1	N
Static Front Axle Load	1254.2	N
Static Rear Axle Load	1737.8	N
Dynamic Front Axle Load (braking)	1649.4	N
Dynamic Rear Axle Load (braking)	1342.7	N
Maximum Braking Force	2094.4	N
Max Deceleration	6.87	m/s ² (0.70 g)
Front Brake Torque / Wheel	100.4	Nm
Rear Brake Torque / Wheel	139.0	Nm
Effective Disc Radius	66.6	mm
Front Caliper Clamp Force	1753	N
Rear Caliper Clamp Force	2429	N
Front Line Pressure	2209	kPa (22.1 bar)
Rear Line Pressure	3061	kPa (30.6 bar)
Master Cylinder Output Force	2044	N
Brake Bias (front)	42%	—
Required Pedal Force	340.7	N
Theoretical Stopping Distance	80.9	m

Table 4: Summary of Brake Calculation Results

5 CAD Models

5.1 Brake Disc

The brake disc was designed to maximise the swept area for heat dissipation while conforming to the packaging envelope imposed by the 11-inch wheel rim. The rotor geometry was modelled in SolidWorks, incorporating:

- Outer diameter: 185 mm, inner diameter: 120 mm — annular swept face
- Disc thickness: 6.35 mm (0.25 in), compatible with Wilwood GP200 pad width
- Vented/slotted design to enhance thermal dissipation under repeated braking

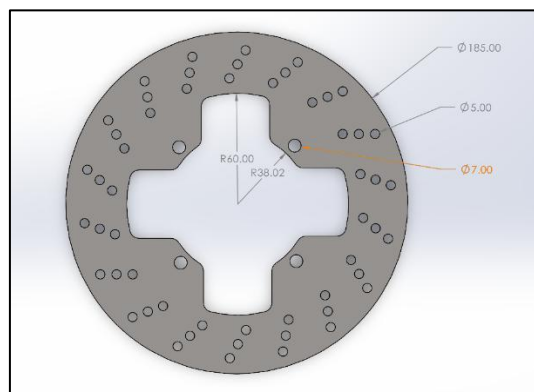


Figure 1 — Brake Disc (Front View)

5.2 Brake Pedal

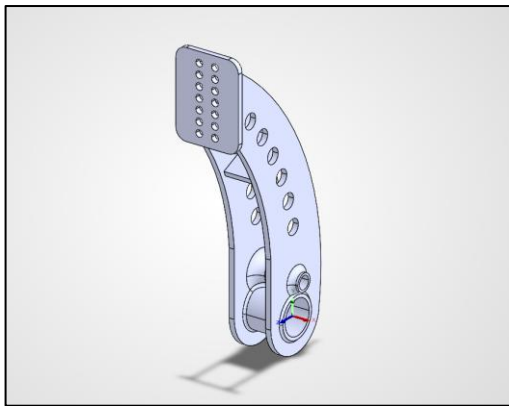


Figure 2 — Brake Pedal (Isometric View)

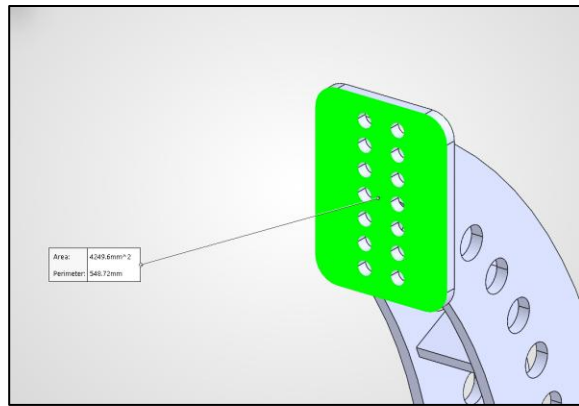


Figure 3 — Brake Pedal Area Selection

5.3 Pedal Box Assembly

The pedal box assembly is constituted of the brake and throttle pedals, the master cylinders, the reservoirs, the bias bar and the necessary mounts and support structures.

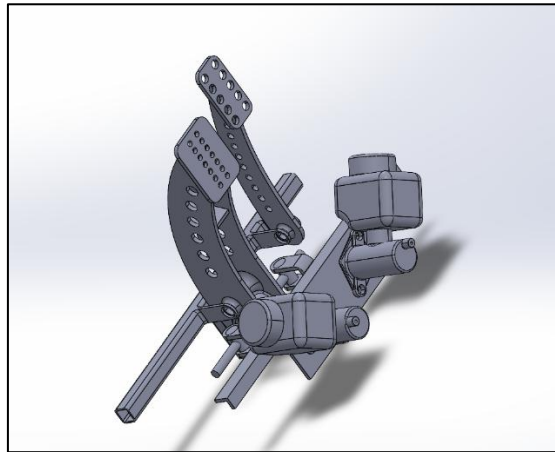


Figure 4 — Pedal Box Assembly

6 FEA: Thermal and Structural Analysis

6.1 Static Structural Analysis of the Brake Discs

Static structural FEA was performed in ANSYS Mechanical on both the front and rear brake disc models to verify that the discs do not yield or fail under the peak caliper clamping force of 2429 N applied at the pad contact area. The disc was constrained at its bolt-hole mounting faces, and the caliper load was applied as a distributed pressure over the pad footprint.

Material chosen: **Stainless Steel** (after comparing results with Grey Cast Iron and Aluminum Alloy)

Key results from the structural analysis:

Front Brake Disc:

- Maximum Von-Mises (Equivalent) Stress: 1.76 MPa
- Maximum Total Deformation: 8.93×10^{-5} mm

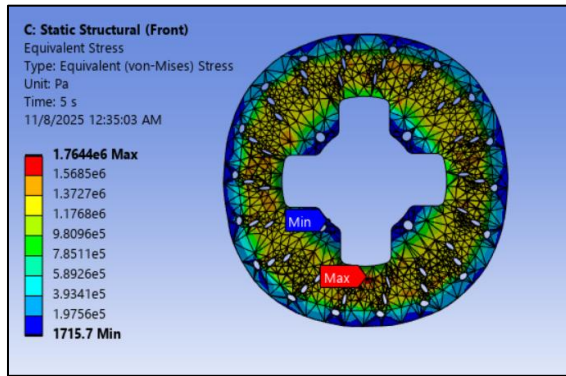


Figure 5 — Von-Mises Stress — Front Brake Disc

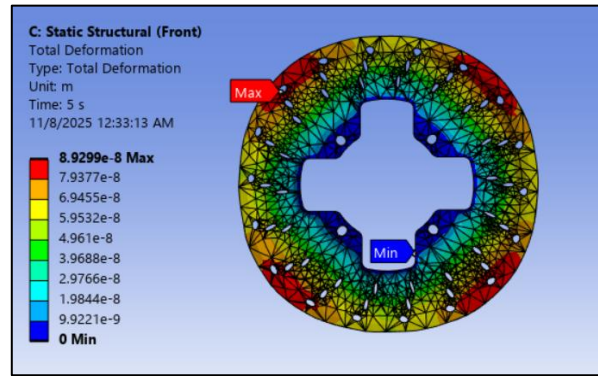


Figure 6 — Total Deformation — Front Brake Disc

Rear Brake Disc:

- Maximum Von-Mises (Equivalent) Stress: 2.44 MPa
- Maximum Total Deformation: 1.24×10^{-4} mm

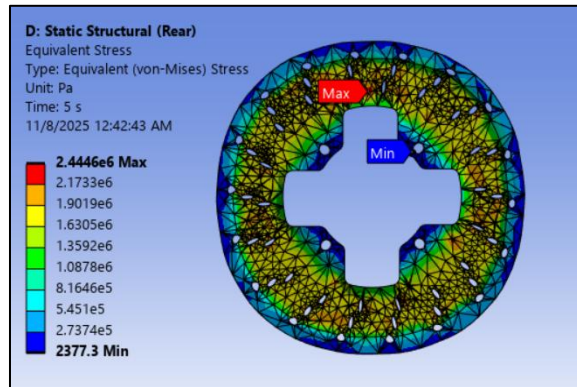


Figure 7 — Von-Mises Stress — Rear Brake Disc

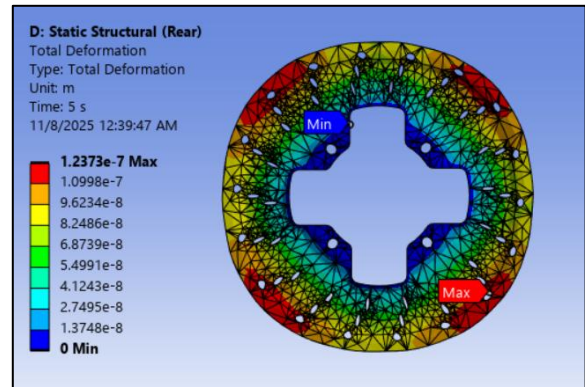


Figure 8 — Total Deformation — Rear Brake Disc

6.2 Transient Thermal Analysis of the Brake Discs

A transient thermal analysis was conducted to evaluate temperature distribution across disc surface during sustained braking. The heat flow input to the disc was estimated from the kinetic energy of the vehicle at maximum speed:

$$KE = \frac{1}{2} \times m \times v_0^2 = \frac{1}{2} \times 305 \times 33.33^2 = 169,410 \text{ J}$$

Calculating the stopping time to be 4.85 seconds and assuming that 90% of kinetic energy is absorbed by the discs:

$$\text{Heat flow (for each disc)} = 0.9 \times KE / (4 \times t) = 0.9 \times 169,410 / (4 \times 4.85) \approx 7,859.23 \text{ W}$$

Boundary conditions: ambient temperature 25 °C, convective cooling from the disc surfaces at forced convection coefficient $h \approx 88.147 \text{ W/m}^2\text{K}$.

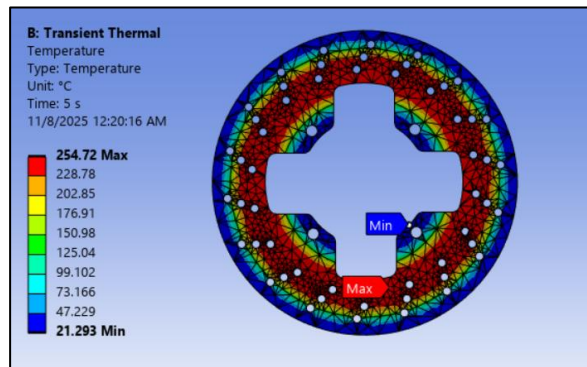


Figure 9 — Temperature Distribution

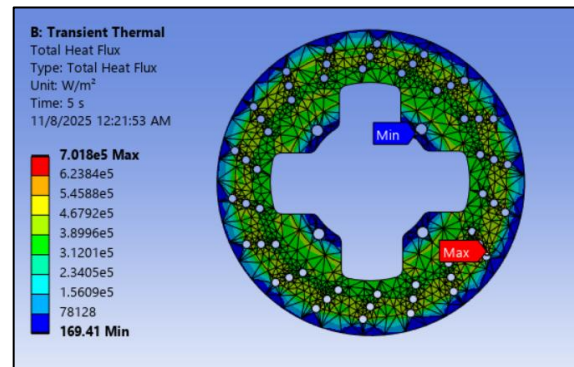


Figure 10 — Total Heat Flux

6.3 Static Structural Analysis of the Brake Pedal

The brake pedal was modelled as a bell-crank lever in Al 6061-T6. A peak driver input of 400 N was applied at the pedal pad contact point. The pivot was constrained, and the master cylinder push-rod connection was modelled as a reaction force boundary. Key outputs included maximum Von-Mises stress and total deformation.

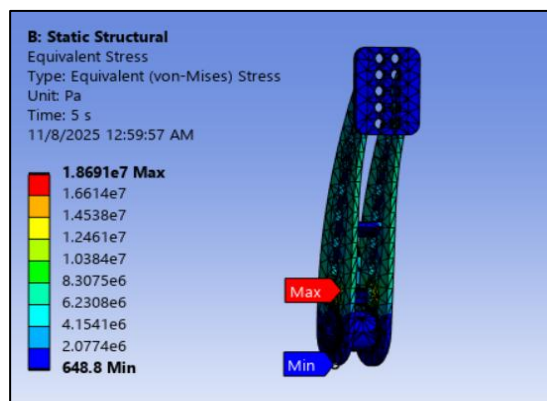


Figure 11 — Von-Mises Stress — Brake Pedal

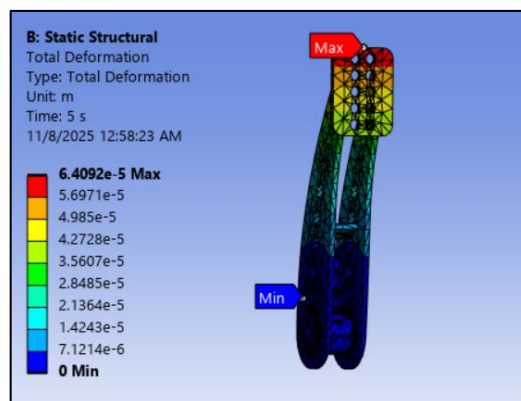


Figure 12 — Total Deformation — Brake Pedal

7 Conclusion

This report has presented the complete design and development workflow for the hydraulic braking system of the IITG Racing Formula Student Electric Vehicle. The following conclusions are drawn:

- The Wilwood Billet GP200 caliper was selected as the optimal choice across all four corners, offering the lightest available option (0.9 lbs) with sufficient piston area (793.547 mm²) for the required braking performance at acceptable line pressures.
- Brake force calculations confirmed that the system achieves a maximum deceleration of 0.70 g, generating front and rear caliper clamp forces of 1753 N and 2429 N respectively, with corresponding line pressures of 22.1 bar (front) and 30.6 bar (rear).
- The required pedal force of 340.7 N (with a 6:1 pedal ratio) is within ergonomic limits for competition drivers, and the theoretical stopping distance from 120 km/h is 80.9 m under ideal conditions.
- The 42% front brake bias provides a conservative starting point for bias bar adjustment on circuit; the dual-circuit layout enables real-time tuning by the driver.
- FEA (structural and thermal) validates that the disc and pedal designs operate within safe stress margins under peak braking loads, with adequate thermal dissipation from the vented disc geometry.

The braking system as designed meets all Formula Bharat and Formula Student technical regulations and provides a robust, lightweight, and tunable foundation for competition. Further development will include detailed testing, bias map generation across multiple circuit configurations, and pad compound selection based on operating temperature data collected during shakedown.

8 References

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